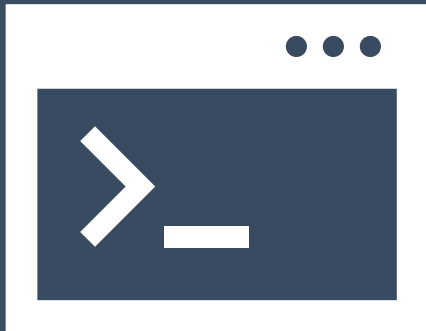


Introduction to Single-cell RNA-seq Analysis

<https://tinyurl.com/hbc-intro-scrna>



Harvard Chan Bioinformatics Core



Introductions!





Shannan Ho Sui
Director



Lorena Pantano
*Director of Bioinformatics
Platform*



John Quackenbush
Faculty Advisor



James Billingsley



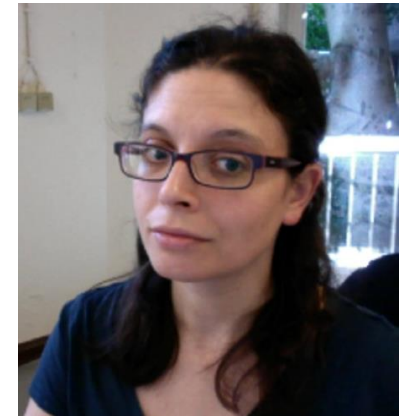
Elizabeth
Partan



Will Gammerdinger



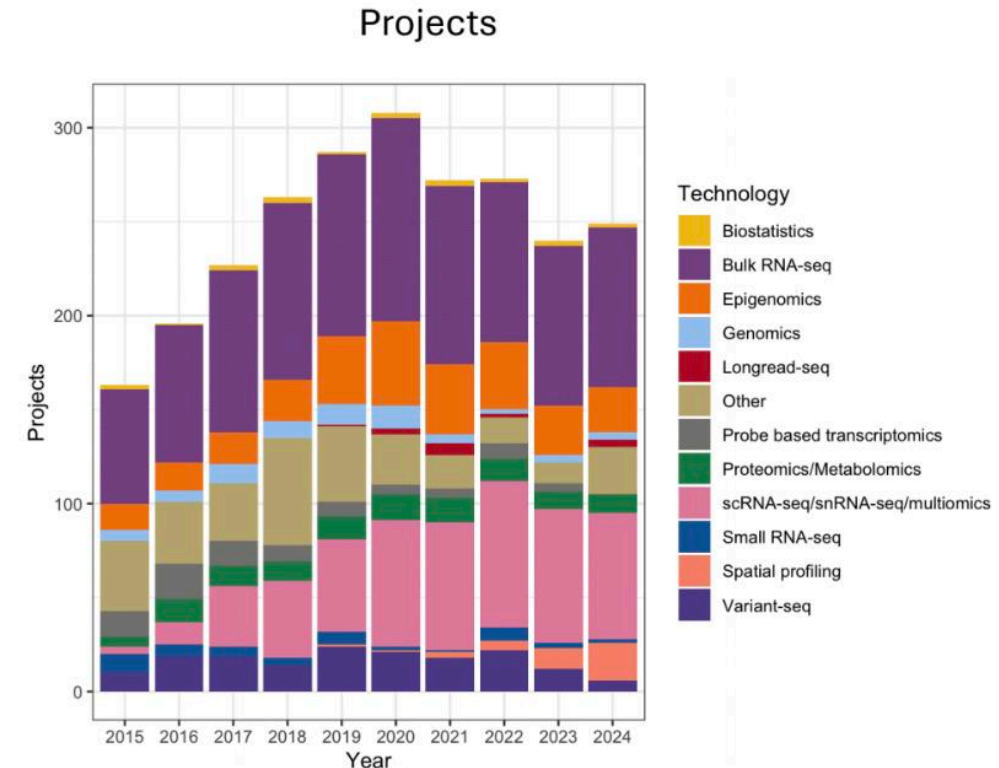
Noor Sohail



Emma Berdan

Consulting

- ❖ Transcriptomics: Bulk, single cell, small RNA
- ❖ Epigenomics: ChIP-seq, CUT&RUN, ATAC-seq, DNA methylation
- ❖ Variant discovery: WGS, resequencing, exome-seq and CNV
- ❖ Multiomics integration
- ❖ Spatial biology
- ❖ Experimental design and grant support



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SCHOOL OF PUBLIC HEALTH

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THE HARVARD CLINICAL
AND TRANSLATIONAL
SCIENCE CENTER



HARVARD
MEDICAL SCHOOL

Training

- ❖ Hands-on workshops design to reflect best practices, reproducibility and an emphasis on experimental design

- ❖ Basic Data Skills

- ❖ Shell

- ❖ R

- ❖ Python

- ❖ Advanced Topics: Analysis of high-throughput sequencing data

- ❖ Chromatin Biology

- ❖ Bulk RNA-seq

- ❖ Differential Gene Expression

- ❖ scRNA-seq

- ❖ Variant Calling

- ❖ Current Topics in Bioinformatics

<https://hsph.harvard.edu/research/bioinformatics/training/>

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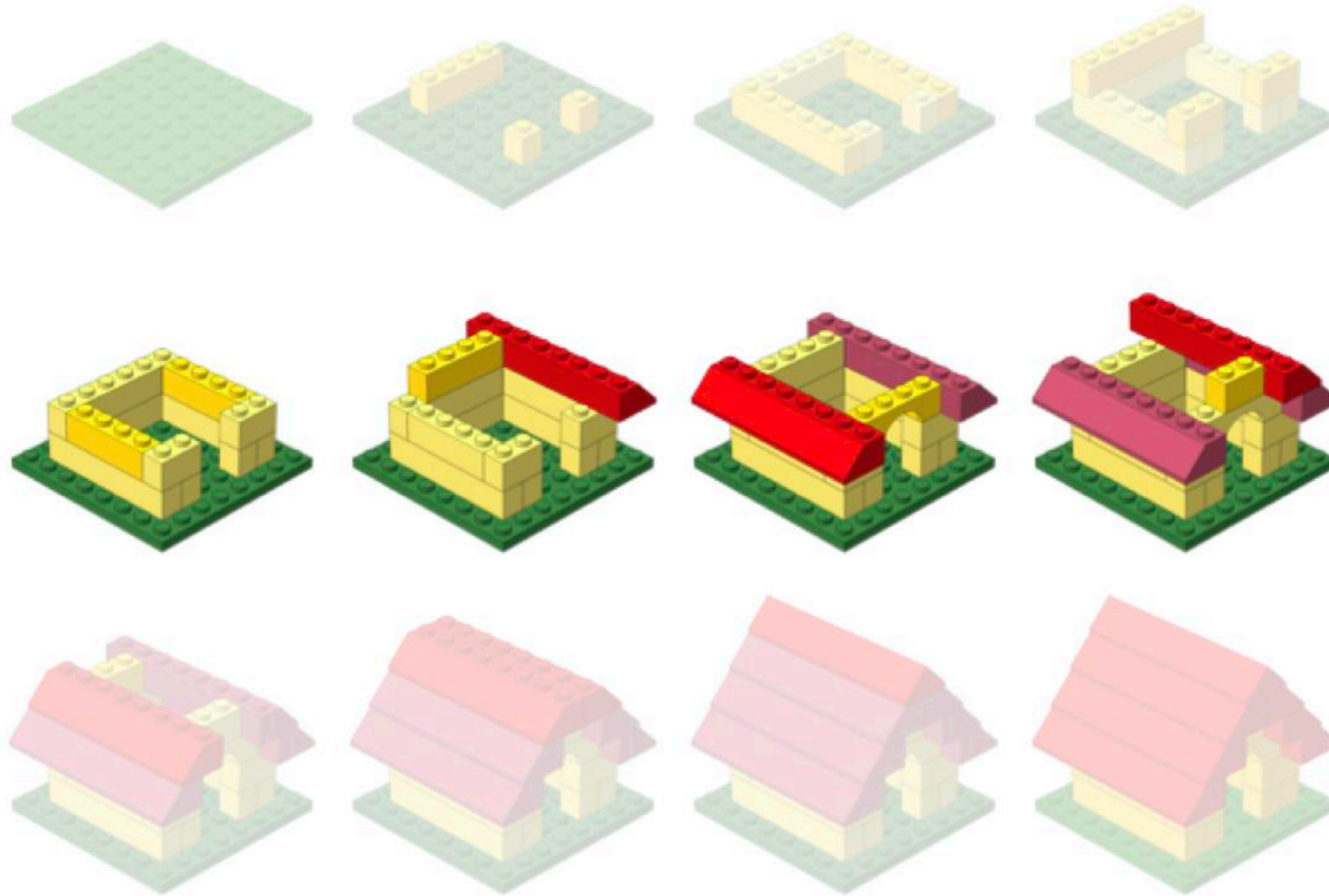
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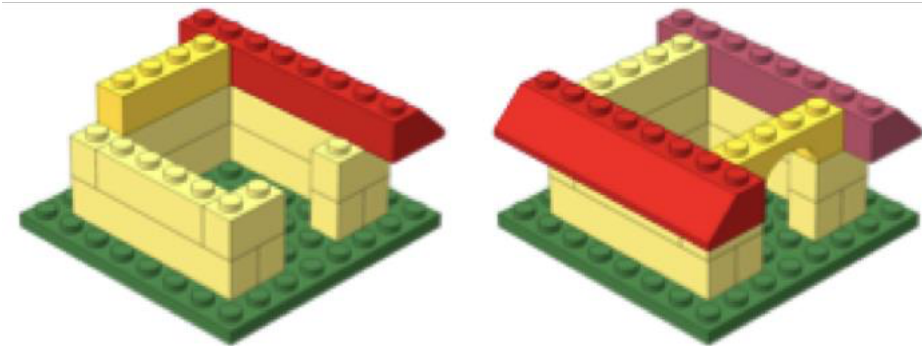


Workshop scope



Bioinformatic Data Analysis

Workshop Scope



- ❖ Describe best practices for designing a single-cell RNA-seq experiment
- ❖ Describe steps in a single-cell RNA-seq analysis workflow
- ❖ Use Seurat and associated tools to perform analysis of single-cell expression data, including data filtering, QC, integration, clustering, and marker identification
- ❖ Understand practical considerations for performing scRNA-seq, rather than in-depth exploration of algorithm theory

Logistics



Course schedule

Day 1

Time	Topic	Instructor
09:30 - 09:45	<u>Workshop introduction</u>	Will
09:45 - 10:35	<u>Introduction to Single Cell RNA-sequencing: a practical guide</u>	<u>Dr. Mandovi Chatterjee</u>
10:35 - 10:40	Break	
10:40 - 11:00	scRNA-seq pre-reading discussion	Noor
11:00 - 11:45	<u>Quality control set-up</u>	Will
11:45 - 12:00	Overview of self-learning materials and homework submission	Will

<https://tinyurl.com/hbc-intro-scrna>

Course materials

❖ We continuously update our materials to reflect changes in the field/software

[Introduction to single-cell RNA-seq](#) [Schedule](#) [HBC](#) [GitHub](#) [Contact us](#) 

Pre-reading: >

Day 1: >

Day 1 Self-learning: >

Day 2: >

Day 2 Self-learning: >

Day 3: >

Day 3 Self-learning: >

Day 4: >

Loading single cell RNA-seq data into Seurat

R PROGRAMMING

SINGLE-CELL RNA-SEQ

QUALITY CONTROL

In this lesson, participants learn how to load single-cell RNA-seq count data using Seurat. The lesson introduces the structure of typical 10X Genomics output files, demonstrates how to import these files efficiently and shows how to generate Seurat objects for one or multiple samples. Participants also practice using loops to streamline data loading and explore the metadata automatically generated by Seurat. By the end, participants will be able to prepare their data for downstream quality control and analysis.

KEYWORDS

R, data import, 10X Genomics

Approximate time: 90 minutes

Learning Objectives:

- Setup R environment for single-cell analysis.
- Establish good data and metadata management.
- Demonstrate how to import data and set up project for upcoming quality control analysis.

On this page

[Learning Objectives:](#)

[Quality control set-up](#)

 Code

Course participation

- ❖ Mandatory review of self-learning lessons and assignments
- ❖ Attendance required for all classes
- ❖ Your questions and active participation drive learning
- ❖ **We look forward to all of your questions!**



Course participation

- ❖ At-home lessons and exercises after each session
- ❖ Cover material not previously discussed
- ❖ Provides us feedback to help pace the course appropriately
- ❖ 3-5 hours to complete

Using AI for Assignments

❖ Do

- ❖ Try to resolve error messages with it
- ❖ Test code written by AI on a dataset where you have expected results
- ❖ Take the time to review the generated code line-by-line

❖ Don't

- ❖ Implement it in replacement to learning
- ❖ Write code that you don't understand
- ❖ Assume the output from an AI process is correct

Odds & Ends

- ❖ Quit/minimize all applications that are not required for class
- ❖ Name tags
- ❖ Post-its
 - ❖  green - I am all set
 - ❖  red - I need time/help
- ❖ Phones on vibrate/silent
- ❖ Bathrooms

Contact Us

- ❖ *HBC training team:* hbctraining@hsph.harvard.edu
- ❖ *HBC consulting:* bioinformatics@hsph.harvard.edu